

Context-Aware Explainable AI Curator for Virtual Museums

Requirements & Implementation Plan

1. Project Overview

Research question: Does retrieval-grounded, confidence-scored generation reduce hallucinated/unverifiable answers in VR museum AI guides compared to an ungrounded LLM?

Core contribution: A RAG + XAI curator, delivered through a VR/AR museum built on real Smithsonian Open Access data, that cites sources, shows confidence, and explicitly refuses out-of-scope questions.

2. Functional Requirements

ID	Requirement	Status
FR1	Ingest museum artifacts with rich descriptions, tags, galleries	✓ Done (15 artifacts)
FR2	Chunk long text into fine-grained retrievable pieces	✓ Done (110 chunks)
FR3	Dense vector retrieval via embeddings + cosine similarity	✓ Done
FR4	Confidence score shown with every answer (authentic, no boosting)	✓ Done
FR5	Hard refusal for out-of-scope questions, before LLM call	✓ Done
FR6	Grounded generation — answer uses only retrieved context	✓ Done
FR7	Tone personalization (educational/concise/friendly)	✓ Done
FR8	Session memory (conversation history, bounded)	✓ Done
FR9	Related-exhibit recommendations	✓ Done
FR10	Dual query routing (active-exhibit vs. museum-wide)	✓ Done
FR11	Tour memory, progress tracking, badges	✓ Done
FR12	FastAPI backend with <code>/ask</code> , <code>/health</code>	✓ Done
FR13	VR gallery + phone AR delivery, frontend fallback	✓ Done
FR14	Ablation controls (disable refusal gate / memory / vary top-k)	✗ Missing
FR15	Structured request logging (CSV/DB of every query + outcome)	✗ Missing
FR16	Baseline system: ungrounded LLM	✗ Missing
FR17	Baseline system: RAG without refusal gate	✗ Missing

FR18 Evaluation harness (run question set across all systems)

✖ Missing

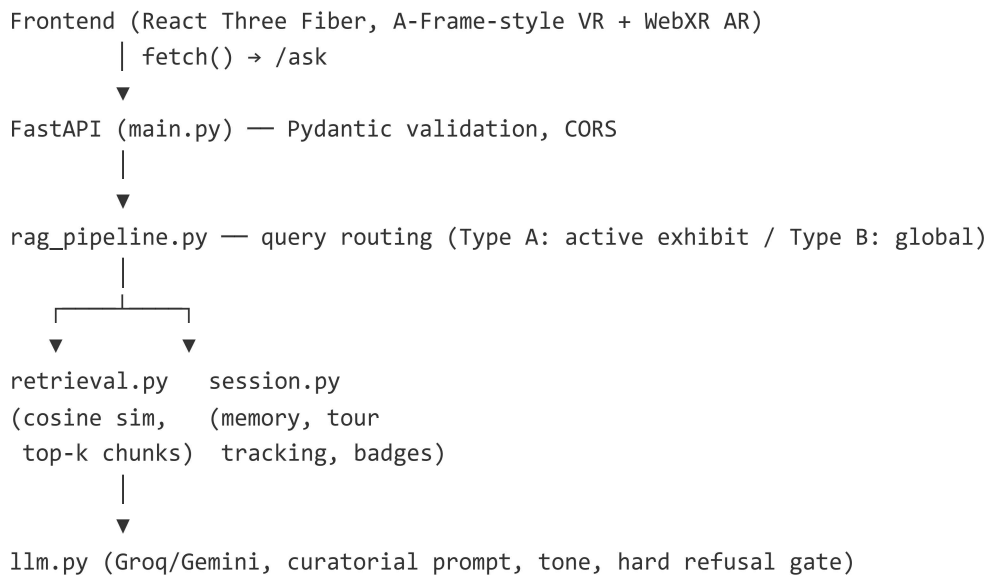
3. Non-Functional Requirements

- **Groundedness:** no answer may include facts absent from retrieved context
- **Determinism of refusal:** below-threshold confidence must never reach the LLM
- **Reproducibility:** fixed random seeds, documented confidence threshold, versioned dataset
- **Latency:** acceptable for a live demo (no hard number set — test on demo hardware)
- **Auditability:** every answer traceable to its source chunk(s) and confidence score

4. Data Requirements

- `artifacts.json`: id, title, gallery, description, significance, funFact, relatedTopics, modelPath, attribution
- 15 artifacts currently; expansion to ~30–45 (10–15/gallery) optional, lower priority than evaluation work
- 110 semantic chunks (2–3 sentences each) — done
- Attribution text pulled from Smithsonian `read.txt` / `read.md` per model

5. Technical Architecture



6. Remaining Implementation Tasks

6.1 Ablation Layer (build first)

- `confidence_threshold_enabled`: bool
- `memory_enabled`: bool

- top_k: int (parameterize, currently likely fixed)
- Expose as function parameters in rag_pipeline.py, not user-facing UI

6.2 Structured Logging (build second — unblocks everything after)

Log per /ask call: timestamp, session_id, question, retrieved_chunks[], scores[], confidence, answer, refusal_fired (bool), config_used → append to CSV or lightweight SQLite table

6.3 Baseline Systems

- **Baseline 1 (ungrounded):** direct LLM call, generic prompt, no retrieval
- **Baseline 2 (RAG, no refusal):** existing pipeline with FR14's refusal flag off
- Wrap all three (baselines + full system) behind one callable interface for the evaluation harness

6.4 Evaluation Harness

- Script: loop a fixed question set × all systems × relevant ablation configs → log via 6.2
- Output: one CSV row per (question, system, config) combination

7. Evaluation Plan

Step	Detail
Question set	20–30 questions: in-scope, ambiguous, out-of-scope — spread across all 3 galleries
Manual annotation	You + guide/teammate independently label: hallucinated? faithful? correctly refused?
Inter-rater agreement	Compute % agreement between the two annotators
Metrics	Refusal accuracy, hallucination rate, faithfulness rate, confidence calibration (bucket score → accuracy)
Ablation results	Compare full system vs. threshold-off vs. memory-off
User study	N=8–10, short Likert survey (trust, source-clarity) comparing baseline vs. full system

8. Research Paper Plan

Structure: Abstract → Introduction (contribution bullets) → Related Work (lit review as prose, grouped by theme) → System Architecture → Implementation → Evaluation → Results & Discussion → Limitations → Conclusion & Future Work → References

Limitations to state explicitly: artifact-set size, single-source dataset (Smithsonian), small user-study N, manual annotation (mitigate with inter-rater check)

Target venue: AIXVR (direct topic match, per Filho et al. in your lit review) or a workshop/extended-abstract

9. Milestones

Week	Deliverable
1	Ablation flags + logging live; every new test question becomes usable data
2	Both baselines built; evaluation harness running
3	Full evaluation run + manual annotation + inter-rater check complete
4	Metrics computed; user study run; results visualized
5	Paper draft (all sections)
6	Guide review, revisions, formatting to venue template, submission

10. Immediate Next Action

Implement **6.1 (ablation flags)** and **6.2 (logging)** — smallest changes, and every test run from this point becomes paper-usable evidence instead of being discarded.

